Electron shells

Student: .... Class: ....

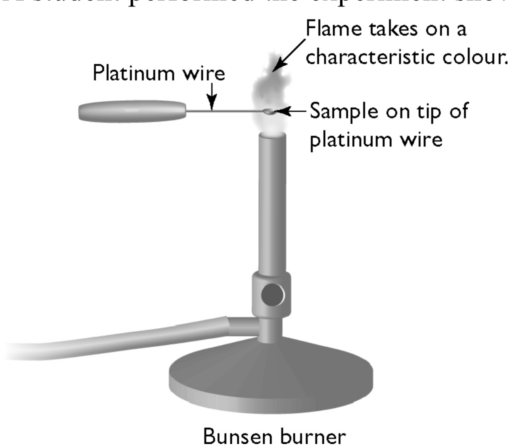
Electron configuration

1. Complete the following table.

Atom or ion	Proton number	Electron number	Electron configuration
Sodium atom (Na)	11		
Fluorine atom (F)		9	
Carbon atom (C)			2, 4
Calcium atom (Ca)			2, 8, 8, 2
Chloride ion (Cl^-)			2, 8, 8
Oxide ion (O^{2-})	8		
Magnesium ion (Mg^{2+})	12		2, 8
Aluminium ion (Al^{3+})	13		

Exciting the electrons

2. A student performed the experiment shown in the diagram below.



A sample of salt (sodium chloride) is placed on the tip of a platinum wire. When the sample is heated in a blue Bunsen flame the flame turns yellow.

Use the theory of electron shells to explain this observation.

....

....

....

....

....

....

....

Electron shell population

3. Each electron shell is represented by a number (*n*) as shown in the following table.

Shell number (*n*)	Maximum electron population
1	
2	
3	
4	

The maximum electron number in each shell is given by the formula:

$$\text{maximum electron number} = 2n^2$$

Complete the table by calculating the maximum electron population in each shell.